Lecture 2. Basic Quantities

Survival, Hazard, Cumulative Hazard, Quantiles, Residual Life

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1 Review: Classifying Incomplete Data

Situation	Type
Clinical trial starts at randomisation (time 0); a participant completes 24 months with no event by study end.	RC
At baseline the tumour is already present; exact onset unknown, only that it occurred before baseline.	LC
Annual screening; disease-free at year 2, first positive at year 3.	IC
Registry enrols patients only if alive at clinic entry (some enter 3 years after diagnosis); followed until death or end of study.	LT + (RC or event)
Historical cohort assembled in 2020 from obituaries of people who died in 2010–2019; age at death recorded, no survivors.	RT

2 Survival Function

Let $X \geq 0$ be a random failure time.

Definition: Survival function

For continuous X with pdf f and cdf $F(x) = P(X \le x)$,

$$S(x) = P(X > x) = 1 - F(x) = \int_x^\infty f(s)\,\mathrm{d}s.$$

$S(x)$ is the probability of surviving beyond time x. It is monotone non-increasing with $S(0) = 1$ and $S(\infty) = 0$.

Graphically, $S(x)$ is the area under the density to the right of x. (The slides show the Exp(0.5) density and the 1989 US all-cause survival curves by race/sex, which stay near 1 until middle age and then drop steeply.)

3 Hazard Function

Definition: Hazard function

$$h(x) = \lim_{\Delta x \to 0} \frac{P(x \le X < x + \Delta x \mid X \ge x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{P(x \le X < x + \Delta x)/P(X \ge x)}{\Delta x} = \frac{f(x)}{S(x)}.$$

- $f(x)\,\mathrm{d}x \approx P(x \le X < x + \mathrm{d}x) = S(x) - S(x + \mathrm{d}x)$ is the "infinitesimal" probability of failure at x, *unconditionally* on being alive before x.
- $h(x)\,\mathrm{d}x$ is the infinitesimal probability of failure in the next instant after x, *given* alive at x.

Remark: h is a rate, not a probability

$h(x) \ge 0$ can exceed 1; only $h(x)\Delta x$ for small Δx approximates a conditional probability.

3.1 Shapes of hazard functions

(i) **Constant**: e.g. the exponential distribution.

(ii) **Increasing**: industrial systems and cars, whose component failure rates rise over time (ageing).

(iii) **Decreasing** (less common): e.g. transplant studies (risk is highest just after the operation).

(iv) **Bathtub-shaped**: population mortality (high infant mortality, low in mid-life, rising in old age).

(v) **Hump-shaped (unimodal)**: rises early and eventually declines; e.g. breast cancer treatment studies where surgery- or toxicity-related high-risk patients tend to die early.

3.2 Examples

Example: Exponential(λ)

$f(x) = \lambda e^{-\lambda x}$ and

$$S(x) = \int_x^\infty \lambda e^{-\lambda s}\,\mathrm{d}s = \big[-e^{-\lambda s}\big]_x^\infty = e^{-\lambda x}, \qquad h(x) = \frac{f(x)}{S(x)} = \lambda.$$

The constant hazard is the *memoryless* property: $P(X > x + t \mid X > x) = e^{-\lambda t}$ does not depend on x.

Example: Weibull(α, λ)

$S(x) = \exp(-\lambda x^\alpha)$. Then

$$f(x) = -\frac{\mathrm{d}S(x)}{\mathrm{d}x} = \alpha\lambda x^{\alpha-1}\exp(-\lambda x^\alpha), \qquad h(x) = \frac{f(x)}{S(x)} = \alpha\lambda x^{\alpha-1}.$$

$\alpha > 1$: increasing hazard; $\alpha = 1$: constant (exponential); $\alpha < 1$: decreasing.

4 Cumulative Hazard and the S–h Relationship

Theorem: Continuous case

Define the **cumulative hazard** $H(x) = \int_0^x h(s)\,\mathrm{d}s$. If X is absolutely continuous,

$$H(x) = -\ln S(x), \qquad S(x) = \exp\{-H(x)\} = \exp\Big\{-\int_0^x h(s)\,\mathrm{d}s\Big\}.$$

Proof. Since $f(s) = -S'(s)$,

$$H(x) = \int_0^x \frac{f(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \frac{S'(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \mathrm{d}\{\ln S(s)\} = -\big[\ln S(x) - \ln S(0)\big] = -\ln S(x),$$

using $S(0) = 1$. Exponentiating gives $S(x) = e^{-H(x)}$. ∎

Summary of relations (continuous X)	
$f(t) = \dfrac{\mathrm{d}F(t)}{\mathrm{d}t} = -S'(t)$	$F(t) = 1 - S(t)$
$h(t) = \dfrac{f(t)}{S(t)} = -\dfrac{\mathrm{d}}{\mathrm{d}t}\ln S(t)$	$H(t) = \displaystyle\int_0^t h(s)\,\mathrm{d}s = -\ln S(t)$
$S(t) = \exp\{-H(t)\}$	$f(t) = h(t)\exp\{-H(t)\}$

Any one of f, F, S, h, H determines all the others.

5 Discrete Failure Times

At points $0 = x_0 < x_1 < \cdots < x_d$ let $P(X = x_j) = p_j > 0$, $\sum_{j=1}^{d} p_j = 1$. Discrete times arise from rounding of measurements or grouping failure times into intervals. Then

$$S(x) = P(X > x) = \sum_{j=1}^{d} P(X = x_j)\, I(x_j > x),$$

a right-continuous step function.

Definition: Discrete hazard

$$h(x_j) = P(X = x_j \mid X \geq x_j) = \frac{p_j}{S(x_{j-1})} = \frac{p_j}{\sum_{k=j}^{d} p_k}, \qquad j = 1, \ldots, d.$$

Here $h(x_j)$ *is* a conditional probability, so $0 < h(x_j) \leq 1$.

Since $p_j = S(x_{j-1}) - S(x_j)$,

$$h(x_j) = \frac{S(x_{j-1}) - S(x_j)}{S(x_{j-1})} = 1 - \frac{S(x_j)}{S(x_{j-1})}.$$

Example: $p_j = 1/3$, $j = 1, 2, 3$

$$S(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 2/3, & 1 \leq x < 2 \\ 1/3, & 2 \leq x < 3 \\ 0, & x \geq 3 \end{cases} \qquad h(1) = \frac{1/3}{1} = \frac{1}{3}, \quad h(2) = \frac{1/3}{2/3} = \frac{1}{2}, \quad h(3) = \frac{1/3}{1/3} = 1.$$

The last hazard equals 1: anyone alive just before the last support point must fail there.

Theorem: Product-limit representation

$$S(x_j) = \frac{S(x_j)}{S(x_{j-1})} \frac{S(x_{j-1})}{S(x_{j-2})} \cdots \frac{S(x_1)}{S(x_0)} S(x_0) = \prod_{i: x_i \leq x_j} \frac{S(x_i)}{S(x_{i-1})} = \prod_{i: x_i \leq x_j} [1 - h(x_i)],$$

using the telescoping product and $S(x_0) = S(0) = 1$.

This product form is the population version of the Kaplan–Meier estimator (Lecture 5).

Remark: Two definitions of the discrete cumulative hazard

Klein & Moeschberger define $H(x_j) = \sum_{i: x_i \leq x_j} h(x_i)$. With this definition $S(x) = \exp\{-H(x)\}$ does *not* hold. Cox & Oakes (1984) instead define

$$H(x_j) = -\sum_{i: x_i \leq x_j} \log\left[1 - h(x_i)\right] \approx \sum_{i: x_i \leq x_j} h(x_i),$$

which restores $S = e^{-H}$ exactly (take logs of the product-limit form). The approximation uses $-\log(1 - u) \approx u$ for small u.

6 Mean and Mean Residual Life

Theorem: Mean as area under S

For $X \geq 0$, $\mathrm{E}(X) = \int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty S(t)\,\mathrm{d}t$.

Proof. Write $t = \int_0^t \mathrm{d}u$ and exchange the order of integration (Tonelli, integrand ≥ 0):

$$\int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty \int_0^t f(t)\,\mathrm{d}u\,\mathrm{d}t = \int_0^\infty \int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_0^\infty S(u)\,\mathrm{d}u.$$

■

Definition: Mean residual life

$\mathrm{mrl}(x) = \mathrm{E}(X - x \mid X > x)$. For continuous X,

$$\mathrm{mrl}(x) = \frac{\int_x^\infty (t-x) f(t)\,\mathrm{d}t}{S(x)} = \frac{\int_x^\infty S(t)\,\mathrm{d}t}{S(x)},$$

the area under the survival curve to the right of x divided by $S(x)$. Note $\mathrm{mrl}(0) = \mathrm{E}(X)$.

Proof of the second equality. Same Fubini argument with $t - x = \int_x^t \mathrm{d}u$: $\int_x^\infty (t-x) f(t)\,\mathrm{d}t = \int_x^\infty \int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_x^\infty S(u)\,\mathrm{d}u$. ■

7 Quantiles

Definition: p-th quantile

The p-th quantile ($100p$-th percentile) x_p satisfies $S(x_p) \leq 1 - p$:

$$x_p = \inf\{t : S(t) \leq 1 - p\}.$$

For continuous strictly decreasing S this is the unique solution of $S(x_p) = 1-p$, i.e. $F(x_p) = p$. The median is $x_{0.5}$.

Example 1: Exponential(λ) — mean, mrl, median

$S(x) = e^{-\lambda x}$.

- $\mathrm{E}(X) = \int_0^\infty e^{-\lambda t}\,\mathrm{d}t = 1/\lambda$.
- $\mathrm{mrl}(x) = \dfrac{\int_x^\infty e^{-\lambda t}\,\mathrm{d}t}{e^{-\lambda x}} = \dfrac{e^{-\lambda x}/\lambda}{e^{-\lambda x}} = \dfrac{1}{\lambda}$ (constant: memorylessness).
- Median: $e^{-\lambda x_{0.5}} = 1/2 \Rightarrow x_{0.5} = (\ln 2)/\lambda$.

Example: Exercise 1 — Weibull quantile

$S(x) = \exp(-\lambda x^\alpha)$, so $F(x_p) = p$ gives

$$1 - e^{-\lambda x_p^\alpha} = p \implies e^{-\lambda x_p^\alpha} = 1 - p \implies \lambda x_p^\alpha = -\ln(1-p) \implies x_p = \left[\frac{-\ln(1-p)}{\lambda}\right]^{1/\alpha}.$$

In particular the median is $x_{0.5} = (\ln 2/\lambda)^{1/\alpha}$.

Example: Exercise 2 — LogNormal quantile

$X \sim \text{LogNormal}(\mu, \sigma^2)$ means $Y = \ln X \sim N(\mu, \sigma^2)$. Since ln is strictly increasing, quantiles map through it:

$$p = P(X \le x_p) = P(Y \le \ln x_p) = \Phi\Big(\frac{\ln x_p - \mu}{\sigma}\Big) \implies x_p = \exp\left(\mu + \sigma\Phi^{-1}(p)\right),$$

where Φ^{-1} is the standard normal quantile function. With $p = 1/2$, $\Phi^{-1}(1/2) = 0$, so $\text{median}(X) = e^{\mu}$.

8 Median Residual Life

Definition: Median residual life

Preferred to the mrl when the distribution is highly skewed.

$$\text{mdrl}(x) = \text{median}(X - x \mid X > x),$$

equivalently

$$P(X - x > \text{mdrl}(x) \mid X > x) = \frac{1}{2} \iff \frac{P(X > x + \text{mdrl}(x),\ X > x)}{P(X > x)} = \frac{1}{2}.$$

For continuous X the numerator is $S(x + \text{mdrl}(x))$, hence

$$S(x + \text{mdrl}(x)) = \tfrac{1}{2}S(x) \quad \implies \quad \text{mdrl}(x) = S^{-1}\Big(\frac{S(x)}{2}\Big) - x.$$

Example: Exponential(λ)

$S^{-1}(y) = -(1/\lambda)\ln y$ for $y \in (0, 1]$, so

$$\text{mdrl}(x) = -\frac{1}{\lambda}\ln\Big(\frac{e^{-\lambda x}}{2}\Big) - x = -\frac{1}{\lambda}(-\lambda x - \ln 2) - x = \frac{\ln 2}{\lambda}.$$

Again constant in x.

9 Extra Exercise 1: Weibull Mean Residual Life

Let $S(t) = \exp(-\lambda t^{\alpha})$, $\lambda, \alpha > 0$, and $m(t) = \text{E}[T - t \mid T > t]$.

1. Derive $m(t)$. 2. Show $m(t)$ is constant when $\alpha = 1$. 3. For $\alpha > 1$, show $m(t)$ is decreasing and interpret.

Derivation: substitution $v = \lambda u^{\alpha}$

With $f(u) = \lambda\alpha u^{\alpha-1}e^{-\lambda u^{\alpha}}$, the numerator is

$$N(t) = \int_t^{\infty} (u - t)\,\lambda\alpha u^{\alpha-1}e^{-\lambda u^{\alpha}}\,\mathrm{d}u.$$

Set $v = \lambda u^\alpha$, so $\mathrm{d}v = \lambda\alpha u^{\alpha-1}\,\mathrm{d}u$ and $u = (v/\lambda)^{1/\alpha}$; limits $u = t \mapsto v = \lambda t^\alpha$, $u \to \infty \mapsto v \to \infty$. Then

$$\begin{aligned} N(t) &= \int_{\lambda t^\alpha}^{\infty} \Big(\big(\tfrac{v}{\lambda}\big)^{1/\alpha} - t \Big) e^{-v}\,\mathrm{d}v = \lambda^{-1/\alpha} \int_{\lambda t^\alpha}^{\infty} v^{1/\alpha} e^{-v}\,\mathrm{d}v - t\int_{\lambda t^\alpha}^{\infty} e^{-v}\,\mathrm{d}v \\ &= \lambda^{-1/\alpha}\,\Gamma\big(1+\tfrac{1}{\alpha}, \lambda t^\alpha\big) - t\,e^{-\lambda t^\alpha}, \end{aligned}$$

where $\Gamma(s,x) = \int_x^\infty v^{s-1}e^{-v}\,\mathrm{d}v$ is the upper incomplete gamma function. Dividing by $S(t) = e^{-\lambda t^\alpha}$:

$$m(t) = \frac{N(t)}{S(t)} = \frac{1}{S(t)} \cdot \frac{1}{\lambda^{1/\alpha}}\,\Gamma\big(1+\tfrac{1}{\alpha}, \lambda t^\alpha\big) - t.$$

- $\alpha = 1$: $\Gamma(2, \lambda t) = (1+\lambda t)e^{-\lambda t}$, so $m(t) = \frac{(1+\lambda t)}{\lambda} - t = \frac{1}{\lambda}$ (constant).
- $\alpha > 1$: the hazard $h(t) = \lambda\alpha t^{\alpha-1}$ is increasing, hence $m(t)$ decreases in t (proved in Extra Exercise 2). Interpretation: the system *ages* — the longer it has survived, the shorter its expected remaining life.

10 Extra Exercise 2: mrl and Hazard Monotonicity

Theorem

Let $T \geq 0$ have survival S, density f, hazard $h = f/S$, and $m(t) = \mathrm{E}[T - t \mid T > t]$. Then

1. $m'(t) = h(t)m(t) - 1$;
2. if h is increasing then m is decreasing; if h is decreasing then m is increasing.

(1) Differential equation. $m(t) = \int_t^\infty S(u)\,\mathrm{d}u / S(t)$. By the quotient rule and $\frac{\mathrm{d}}{\mathrm{d}t}\int_t^\infty S(u)\,\mathrm{d}u = -S(t)$,

$$m'(t) = \frac{-S(t)\cdot S(t) - \big(\int_t^\infty S(u)\,\mathrm{d}u\big) S'(t)}{S(t)^2}.$$

Since $S'(t) = -f(t) = -h(t)S(t)$,

$$m'(t) = \frac{-S(t)^2 + \big(\int_t^\infty S(u)\,\mathrm{d}u\big) h(t) S(t)}{S(t)^2} = -1 + h(t)\frac{\int_t^\infty S(u)\,\mathrm{d}u}{S(t)} = -1 + h(t)m(t).$$

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(2) Monotonicity. By (1), $m'(t) \leq 0 \iff h(t)m(t) \leq 1$ and $m'(t) \geq 0 \iff h(t)m(t) \geq 1$. Substituting $u \mapsto t+u$ and using $S(t+u)/S(t) = \exp\{-(H(t+u) - H(t))\}$,

$$m(t) = \int_0^\infty \frac{S(t+u)}{S(t)}\,\mathrm{d}u = \int_0^\infty \exp\Big(-\int_t^{t+u} h(s)\,\mathrm{d}s\Big)\,\mathrm{d}u.$$

- If h is increasing, $h(s) \geq h(t)$ on $[t, t+u]$, so $\int_t^{t+u} h(s)\,\mathrm{d}s \geq u\,h(t)$ and

$$m(t) \leq \int_0^\infty e^{-uh(t)}\,\mathrm{d}u = \frac{1}{h(t)} \;\Rightarrow\; h(t)m(t) \leq 1 \;\Rightarrow\; m'(t) \leq 0.$$

- If h is decreasing, $\int_t^{t+u} h(s)\,\mathrm{d}s \leq u\,h(t)$, so $m(t) \geq 1/h(t)$, $h(t)m(t) \geq 1$, and $m'(t) \geq 0$.

■

Example: Application to the Weibull

$h(t) = \lambda\alpha t^{\alpha-1}$.

- $\alpha < 1$: h decreasing $\Rightarrow$ m increasing.
- $\alpha = 1$: $h \equiv \lambda \Rightarrow m'(t) = \lambda m(t) - 1$; the bounded solution is $m \equiv 1/\lambda$ (so $m' = 0$).
- $\alpha > 1$: h increasing $\Rightarrow$ m decreasing.